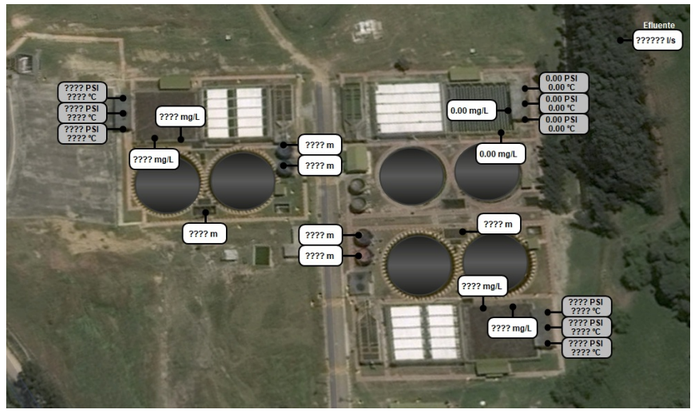
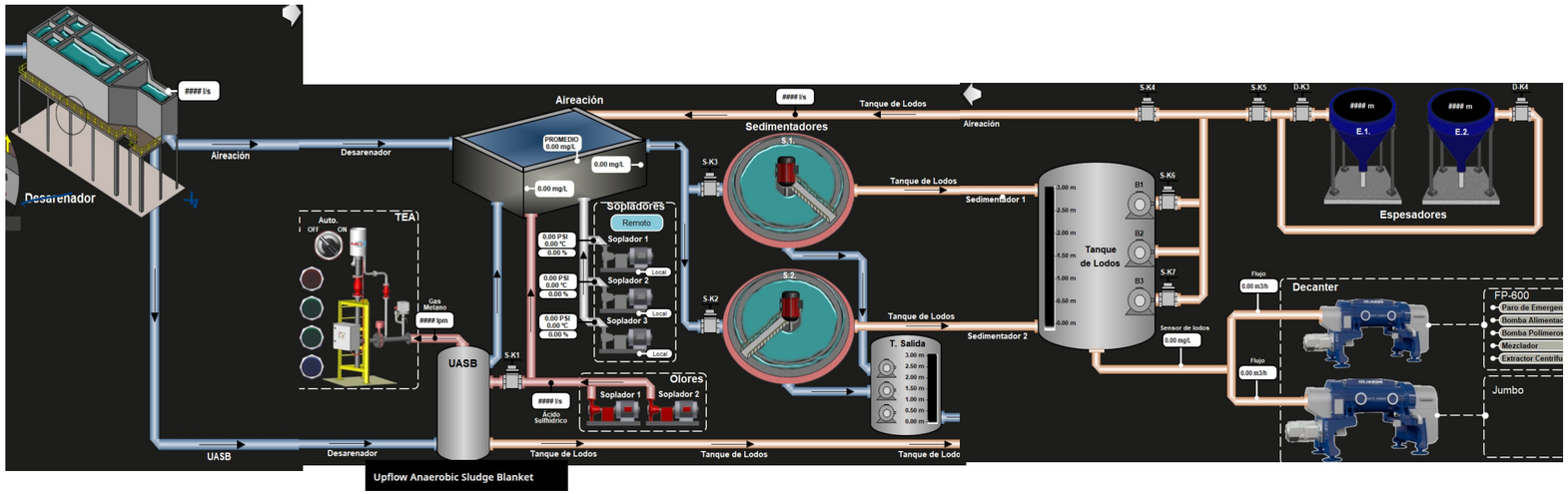
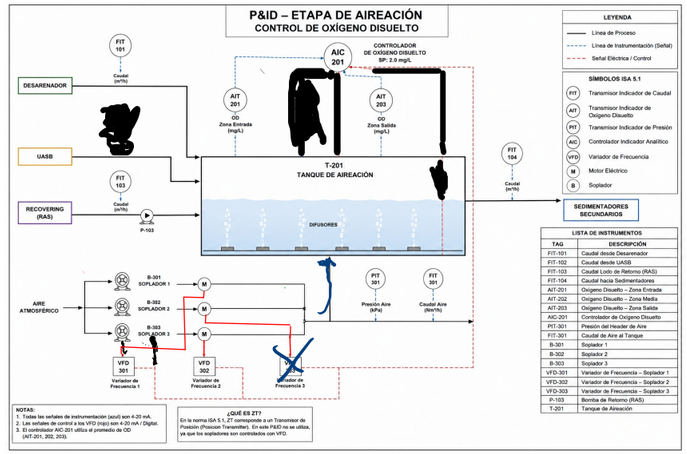




Upflow Anaerobic Sludge Blanket



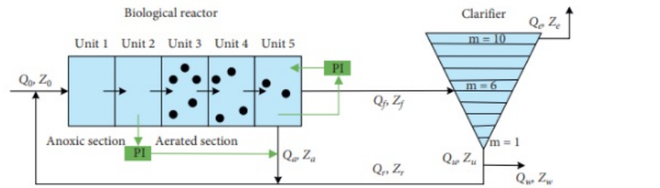


FIGURE 1: Layout of the BSM1.

ADRC Basado En observ

the plant Ecuation :

$$\frac{dS_{O,5}}{dt} = \frac{1}{V_5}(Q_4 S_{O,4} + r_{Z,5} + K_{La5} V_5 (S_O^* - S_{O,5}) - Q_5 S_{O,5}) \quad (1)$$

$$r_{Z,5} = -\frac{1-Y_H}{Y_H}\rho_1 - \frac{4.57-Y_A}{Y_A}\rho_3 \quad (2)$$

$$\rho_1 = \mu_H\left(\frac{S_S}{K_S+S_S}\right)\left(\frac{S_O}{K_{O,H}+S_O}\right)X_{B,H} \quad (3)$$

$$\rho_3 = \mu_A\left(\frac{S_{NH}}{K_{NH}+S_{NH}}\right)\left(\frac{S_O}{K_{O,A}+S_O}\right)X_{B,A} \quad (4)$$

$\frac{dS}{dt} = \frac{Q_{in}}{V} S_{in} + K_{La}(S^* - S) + H_S$

→ oxygen transfer rate

3.1. *Dynamics of the DO Concentration.* In the BSM1, DO concentration of unit 5, i.e., $S_{O,5}$, can be described as [21, 24]

$$\dot{S}_{O,5} = \frac{Q_0 + Q_r + Q_a}{V_4} S_{O,4} + r_5 + K_{La,5}(S_O^* - S_{O,5}) - \frac{Q_0 + Q_r + Q_a}{V_5} S_{O,5}, \quad (1)$$

Ecuación completa (balance de masa real)

$$\frac{dDO}{dt} = \frac{Q_{in}}{V}(DO_{in} - DO) + K_L a(DO_{sat} - DO) - \frac{1}{Y_O}\mu X$$

→ K_{La} variadores

$u = K_{La}$

5. Dinámica de oxígeno disuelto (clave en aireación)

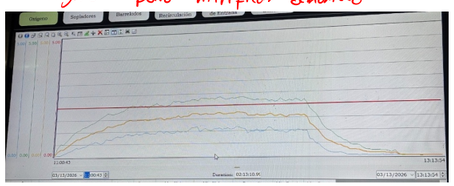
$$\frac{dDO}{dt} = K_L a(DO_{sat} - DO) - \frac{1}{Y_O}\mu X$$

Donde:

- DO: oxígeno disuelto
- DO_{sat}: saturación
- $K_L a$: coeficiente de transferencia de oxígeno (MUY importante para control)
- Y_O: rendimiento respecto al oxígeno

Termino biológico

- Consumen Oxigeno para degradar materia
- Consumen oxigeno para nitrificar amoniaco



BSM1

u → GS → y

The BSM1 can be divided into two parts. One is a biochemical reactor, which is composed of a five-compartment activated sludge reactor consisting of two anoxic tanks ($V_1 = V_2 = 1000\,m^3$) followed by three aerobic tanks ($V_3 = V_4 = V_5 = 1333\,m^3$). The other part is a secondary clarifier ($V = 6000\,m^3$). It is divided into 10 layers evenly. Top layer outputs the water that satisfies the discharge standard, and the bottom layer outflows the waste.

The DO concentration in the fifth tank directly affects the effluent quality, and it is critical to regulate it at an appropriate level. Dynamics of the DO concentration in the fifth tank is[22]

$$\frac{dS_{O,5}}{dt} = \frac{1}{V_5}(Q_4 S_{O,4} + r_{Z,5} + K_{La5} V_5 (S_O^* - S_{O,5}) - Q_5 S_{O,5}) \quad (1)$$

where Q_4 is the effluent flow rate of the fourth biological reactor, $S_{O,4}$ denotes the DO concentration of the fourth biological reactor, K_{La5} is the oxygen transfer coefficient of the fifth biological reactor, S_O^* is the saturation concentration of DO ($8\,g\,m^{-3}$), Q_5 is the effluent flow rate of the fifth biological reactor, $S_{O,5}$ is the DO concentration of the fifth biological reactor, and $r_{Z,5}$ is the appropriate conversion rate of the fifth biological reactor[23]

$$r_{Z,5} = -\frac{1-Y_H}{Y_H}\rho_1 - \frac{4.57-Y_A}{Y_A}\rho_3 \quad (2)$$

$$\rho_1 = \mu_H\left(\frac{S_S}{K_S+S_S}\right)\left(\frac{S_O}{K_{O,H}+S_O}\right)X_{B,H} \quad (3)$$

$$\rho_3 = \mu_A\left(\frac{S_{NH}}{K_{NH}+S_{NH}}\right)\left(\frac{S_O}{K_{O,A}+S_O}\right)X_{B,A} \quad (4)$$

where $X_{B,A}$ and $X_{B,H}$ represent the active autotrophic biomass and the active heterotrophic biomass of the fifth biological reactor, respectively; S_S is the readily biodegradable substrate of the fifth biological reactor; and S_{NH} denotes the NH_4^+ and NH_3 nitrogen of the

/ 9 | The other parameters are listed

Table 1: List of variables in BSM1.

Q_0	Inlet flow rate
Z_0	Influent constituent concentration
Q_a	Internal flow recirculation rate
Z_a	Internal flow constituent concentration
Q_r	Return sludge flow rate
Z_r	Return sludge constituent concentration
Q_f	Flow rate to the secondary clarifier
Z_f	Constituent concentration of the wastewater to the secondary clarifier
Q_u	Underflow rate of the secondary clarifier
Z_u	Constituent concentration of the underflow rate of the secondary clarifier
Q_w	Wastage flow rate
Z_w	Wastage constituent concentration
Q_e	Effluent flow rate
Z_e	Effluent constituent concentration
$S_{O,i}$	Oxygen concentration at compartment i
$K_{La,i}$	Oxygen transfer coefficient at compartment i
AE	Aeration energy
PE	Pumping energy
SP	Sludge production to be disposed
ME	Mixing energy
EC	Consumption of external carbon source
TSS	Suspended solid concentration
V_i	Volume of the tank i
$X_{S,i}$	Slowly biodegradable substrate concentration at tank i
$X_{I,i}$	Particulate inert organic matter concentration at tank i
$X_{B,H,i}$	Active heterotrophic biomass concentration at tank i
$X_{B,A,i}$	Active autotrophic biomass concentration at tank i
$X_{P,i}$	Particulate products arising from biomass decay concemtration at tank i
r_5	Appropriate conversion rate

The list of state variables, with their definition and appropriate notation, is given in Table 1.

Definition	Notation
Soluble inert organic matter	S_I
Readily biodegradable substrate	S_S
Particulate inert organic matter	X_I
Slowly biodegradable substrate	X_S
Active heterotrophic biomass	$X_{B,H}$
Active autotrophic biomass	$X_{B,A}$
Particulate products arising from biomass decay	X_P
Oxygen	S_O
Nitrate and nitrite nitrogen	S_{NO}
$NH_4^+ + NH_3$ nitrogen	S_{NH}
Soluble biodegradable organic nitrogen	S_{ND}
Particulate biodegradable organic nitrogen	X_{ND}
Alkalinity	S_{ALK}

Eight basic processes are used to describe the biological behavior of the system.

- $j = 1$: Aerobic growth of heterotrophs

$$\rho_1 = \mu_H\left(\frac{S_S}{K_S+S_S}\right)\left(\frac{S_O}{K_{O,H}+S_O}\right)X_{B,H}$$

- $j = 2$: Anoxic growth of heterotrophs

$$\rho_2 = \mu_H\left(\frac{S_S}{K_S+S_S}\right)\left(\frac{K_{O,H}}{K_{O,H}+S_O}\right)\left(\frac{S_{NO}}{K_{NO}+S_{NO}}\right)\eta_g X_{B,H}$$

- $j = 3$: Aerobic growth of autotrophs

$$\rho_3 = \mu_A\left(\frac{S_{NH}}{K_{NH}+S_{NH}}\right)\left(\frac{S_O}{K_{O,A}+S_O}\right)X_{B,A}$$

- $j = 4$: Decay of heterotrophs

$$\rho_4 = b_H X_{B,H}$$

- $j = 5$: Decay of autotrophs

$$\rho_5 = b_A X_{B,A}$$

- $j = 6$: Ammonification of soluble organic nitrogen

$$\rho_6 = k_a S_{ND} X_{B,H}$$

- $j = 7$: Hydrolysis of entrapped organics

$$\rho_7 = k_h \frac{X_S/X_{B,H}}{K_X + (X_S/X_{B,H})}\left[\left(\frac{S_O}{K_{O,H}+S_O}\right) + \eta_h\left(\frac{K_{O,H}}{K_{O,H}+S_O}\right)\left(\frac{S_{NO}}{K_{NO}+S_{NO}}\right)\right]X_{B,H}$$

- $j = 8$: Hydrolysis of entrapped organic nitrogen

$$\rho_8 = k_h \frac{X_S/X_{B,H}}{K_X + (X_S/X_{B,H})}\left[\left(\frac{S_O}{K_{O,H}+S_O}\right) + \eta_h\left(\frac{K_{O,H}}{K_{O,H}+S_O}\right)\left(\frac{S_{NO}}{K_{NO}+S_{NO}}\right)\right]X_{B,H}\left(X_{ND}/X_S\right)$$

Table 1. Stoichiometric parameters.

Parameters	Units	Values
Y_A	g cell COD formed (g N oxidized)$^{-1}$	0.24
Y_H	g cell COD formed (g COD oxidized)$^{-1}$	0.67

COD: Chemical Oxygen Demand.

Table 2. Kinetic parameters.

Parameters	Units	Values
μ_H	d^{-1}	4.0
μ_A	d^{-1}	0.5
K_S	$gCOD\,m^{-3}$	10.0
$K_{O,H}$	$g(-COD)\,m^{-3}$	0.2
K_{NH}	$gNH_3\text{-}N\,m^{-3}$	1.0
$K_{O,A}$	$g(-COD)\,m^{-3}$	0.4

Problem statements

The observed conversion rates (r_i) result from combinations of the basic processes:

$r_i = \sum_j \nu_{ij} \rho_j$

- S_I ($i = 1$)

$$r_1 = 0$$

- S_S (i = 2)

$$r_2 = -\frac{1}{Y_H}\rho_1 - \frac{1}{Y_H}\rho_2 + \rho_7$$

- X_I ($i = 3$)

$$r_3 = 0$$

- X_S ($i = 4$)

$$r_4 = (1 - f_P)\rho_4 + (1 - f_P)\rho_5 - \rho_7$$

- $X_{B,H}$ ($i = 5$)

$$r_5 = \rho_1 + \rho_2 - \rho_4$$

- $X_{B,A}$ (i = 6)

$$r_6 = \rho_3 - \rho_5$$

- X_P ($i = 7$)

$$r_7 = f_P\rho_4 + f_P\rho_5$$

- S_O ($i = 8$)

$$r_8 = -\frac{1 - Y_H}{Y_H}\rho_1 - \frac{4.57 - Y_A}{Y_A}\rho_3$$

- S_{NO} ($i = 9$)

$$r_9 = -\frac{1 - Y_H}{2.86\,Y_H}\rho_2 + \frac{1}{Y_A}\rho_3$$

- S_{NH} ($i = 10$)

$$r_{10} = -i_{XB}\rho_1 - i_{XB}\rho_2 - \left(i_{XB} + \frac{1}{Y_A}\right)\rho_3 + \rho_6$$

- S_{ND} ($i = 11$)

$$r_{11} = -\rho_6 + \rho_8$$

- X_{ND} ($i = 12$)

$$r_{12} = (i_{XB} - f_P i_{XP})\rho_4 + (i_{XB} - f_P i_{XP})\rho_5 - \rho_8$$

- S_{ALK} (i = 13)

$$r_{13} = -\frac{i_{XB}}{14}\rho_1 + \left(\frac{1 - Y_H}{14 \cdot 2.86\,Y_H} - \frac{i_{XB}}{14}\right)\rho_2 - \left(\frac{i_{XB}}{14} + \frac{1}{7Y_A}\right)\rho_3 + \frac{1}{14}\rho_6$$

Parameter	Unit	Value
Y_A	g cell COD formed.(g N oxidized)$^{-1}$	0.24
Y_H	g cell COD formed.(g COD oxidized)$^{-1}$	0.67
f_P	dimensionless	0.08
i_{XB}	g N.(g COD)$^{-1}$ in biomass	0.08
i_{XP}	g N.(g COD)$^{-1}$ in particulate products	0.06

Table 2: Stoichiometric parameters

Parameter	Unit	Value
μ_H	d^{-1}	4.0
K_S	g COD.m^{-3}	10.0
$K_{O,H}$	g (-COD).m^{-3}	0.2
K_{NO}	g NO_3-N.m^{-3}	0.5
b_H	d^{-1}	0.3
η_g	dimensionless	0.8
η_h	dimensionless	0.8
k_h	g slowly biodegradable COD.(g cell COD . d)$^{-1}$	3.0
K_X	g slowly biodegradable COD.(g cell COD)$^{-1}$	0.1
μ_A	d^{-1}	0.5
K_{NH}	g NH_3-N.m^{-3}	1.0
b_A	d^{-1}	0.05
$K_{O,A}$	g (-COD).m^{-3}	0.4
k_a	m^3.(g COD . d)$^{-1}$	0.05

Table 3: Kinetic parameters

Control Law

3.1. Dynamics of the DO Concentration. In the BSM1, DO concentration of unit 5, i.e., $S_{O,5}$, can be described as [21, 24]

$$\dot{S}_{O,5} = \frac{Q_0 + Q_r + Q_a}{V_4} S_{O,4} + r_5 + K_{La,5}\left(S_O^* - S_{O,5}\right) - \frac{Q_0 + Q_r + Q_a}{V_5} S_{O,5}, \tag{1}$$

3.2. Control Law Design. Based on the idea of ADRC, system dynamics differs from the integrators connected in series is regarded as the total disturbance. Thus, system (1) can be rewritten as

$$\dot{y} = f + b_0 u, \tag{2}$$

where $y = S_{O,5}$ is the system output, $u = K_{La,5}$ is the control input, b_0 is a tunable parameter, and f is the total disturbance,

$$f = \frac{Q_0 + Q_r + Q_a}{V_4} S_{O,4} + r_5 + K_{La,5}\left(S_O^* - S_{O,5}\right) - \frac{Q_0 + Q_r + Q_a}{V_5} S_{O,5} - b_0 K_{La,5}. \tag{3}$$

3.1. Dynamics of the DO Concentration. In the BSM1, DO concentration of unit 5, i.e., $S_{O,5}$, can be described as [21, 24]

$$\dot{S}_{O,5} = \frac{Q_0 + Q_r + Q_a}{V_4} S_{O,4} + r_5 + K_{La,5}\left(S_O^* - S_{O,5}\right) - \frac{Q_0 + Q_r + Q_a}{V_5} S_{O,5}, \tag{1}$$

$\frac{dSo}{dt} = kLa(S_{Osat} - So) + f_L(S, \dots)$

$[So]$: g/m^3 ; mg/l

setpoint 2 mg/l

$[SO_{sat}] = g/m^3$; mg/l

$\frac{Q}{V} \cdot S = \frac{l/t_n}{l} \cdot \frac{m^3/}{m^3}$

$K_{La,5} = [1/t] \mid K_{La}(S^* - So) : \left[\frac{mg}{l \cdot t}\right]$

$$\frac{dSo}{dt} = KLa\,(So_{sat} - So) + f_1(S_n, So, V, r..)$$

$$\frac{dSo}{dt} = boKLa \underbrace{\left\{ boKLa\,[KLa\,(So_{sat} - So) + f_1] \right\}}_{f}$$

$$\frac{dSo}{dt} = bo\,KLa - f \quad ; \quad \frac{mg}{l \cdot t} = \frac{mg}{l}\frac{1}{t}$$

$$\dot{y} = b_1 u - f$$

Unit:

b_1 →
1. PLC: → 0-100 % → 4-20 mA
2. Velocidad de los motores:

n = 0 - 3000 rpm

3. Caudal de aire

$Q_{aire}\,[V/t] \propto n$

4. Coeficiente de transferencia de oxígeno.

$KLa \propto Q_{aire}$

System: error based:

$$\dot{e_y} = k\,u + \varepsilon$$

Discrete time.

Disturbance one integral.

$$\frac{x_{im}[k+1] - x_{im}(k)}{T} = b_{im}\,e[k]$$

$$x_{im}[k+1] - x_{im}(k) = T\,b_{im}\,e[k]$$

$$x_{im}[k+1] = x_{im}[k] + T\,b_{im}\,e[k]$$

$A_{im} = 1 \qquad b_{im} = T$

Disturbance estimation:

system:

$$x[k+1] = A\,x[k] + b[u[k] + \xi[k]]$$

Disturbance system

$$x_{im}[k+1] = A_{im}\,x_{im}[k]$$
$$\xi[k] = C_{im}\,x_{im}$$

Augment system:

$$x[k+1] = Ax[k] + b\,C_{im}\,x_{im}[k] + bu[k]$$

$$\begin{bmatrix} x[k+1] \\ x_{im}[k+1] \end{bmatrix} = \underbrace{\begin{bmatrix} A & bC_{im} \\ 0 & A_{im} \end{bmatrix}}_{A_{au}} \underbrace{\begin{bmatrix} x[k] \\ x_{im}[k] \end{bmatrix}}_{x_{au}} + \underbrace{\begin{bmatrix} b \\ 0 \end{bmatrix}}_{b_{au}} u$$

$$y = \underbrace{[C\ 0]}_{C_{au}} \begin{bmatrix} x[k] \\ x_{im}[k] \end{bmatrix}$$

The observer:

$$x_{au}[k+1] = A_{au}\,x_{au} + b_{au}\,u[k]$$
$$\hat{x}_{au}[k+1] = A_{au}\,\hat{x}_{au}[k] + b_{au}\,u[k] + L_{au}[e - \hat{e}]$$

$$\hat{x}_{au}[k+1] = [A_{au} - L_{au}C_{au}]\,\hat{x}_{au} + b_{au}\,u[k] - L_{au}\,e[k]$$

$$u = \frac{1}{k}[-v(ey^{(0)}, \xi)]$$

$$u = \frac{1}{k}[-v(\widehat{X}_{au})]$$

$$v = k_1 \cdot X_{au_1} + k_2 \cdot \widehat{X}_{au_2}$$

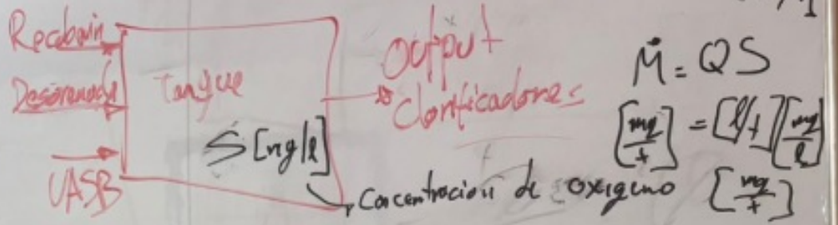
$$v = K \cdot \widehat{X}_{au}$$

$$u = \frac{1}{k}\left[-k_1\,\hat{x}_{au_1}[k] + k_2\,\hat{x}_{au}[k]\right]$$

$$u = \frac{1}{k}\left[-k_1\,\hat{e}_y[k] + k_2\,\hat{\xi}[k]\right]$$

Modelo No lineal:

Caudal Q [l/t] [m^3/t]



$\dot{M} = QS$

$\left[\frac{mg}{t}\right] = \left[\frac{l}{t}\right]\left[\frac{mg}{l}\right]$

$\left[\frac{mg}{t}\right]$

Balance de Masas

Acumulación = Entrada − Salida + Generación − Consumo

$$\frac{dM}{dt} = Q_{i1}S_{i1} + Q_{i2}S_{i2} + Q_{i3}S_{i3} - Q_{out}S + V \cdot K_La(S_0^* - S) - V$$

$$\frac{d[\cdot S]}{dt} = \frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{Q_{out}S}{V} + K_La(S_0^* - S)$$

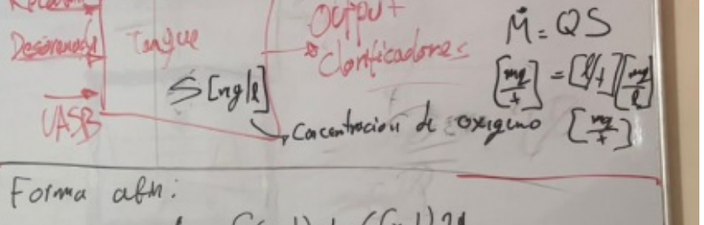
Demostración de K_La:

1. PLC code: 0 − 100%
2. PLC físico: 4 − 20 mA
3. Velocidad del Motor(a) 0 − 60 Hz
   6 → 3000 rpm
4. Caudal de aire: Q_{aire}
5. Coeficiente de transferencia: $K_La \propto Q_{aire}$

$K_La = b_0 u$

Modelo No lineal:

Caudal Q [l/t] [m^3/t]



$\dot{M} = QS$

$\left[\frac{mg}{t}\right] = \left[\frac{l}{t}\right]\left[\frac{mg}{l}\right]$

$\left[\frac{mg}{t}\right]$

Forma afín:

$$\dot{x} = f(x,t) + G(x,t)\cdot u$$

$$\dot{S} = \frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{1}{V}Q_{out}S + K_{La}(S^* - S) - V$$

$$\dot{S} = \frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{1}{V}Q_{out}S - V + K_{La}(S^* - S)$$

$$\dot{S} = f(S,t) + G(S,t)\cdot u$$

Continuar:

Sistema afín:

$$\dot{x} = f(x,t) + G(x,t)\,u$$

$$\dot{S} = \frac{1}{V}\sum_{i=1}^{3} Q_{in}S_{in,i} - \frac{1}{V}Q_{out}S + K_La(S^* - S) - V$$

$$\dot{S} = \frac{1}{V}\sum_{i=1}^{3} Q_{in}S_{in,i} - \frac{1}{V}Q_{out}S - V - K_La\,S + K_La\,S^*$$

$$\dot{S} = \frac{1}{V}\sum_{i=1}^{3} Q_{in}S_{in,i} - \frac{1}{V}Q_{out}S - V - K_La\,S + b_0 u$$

$f(S)$

$$\dot{S} = f(S,t) + b_0 u(t)$$

